

Causal Offset Bounded TE Curve Verification Screen Closeout

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This closeout records the bounded TE Curve Verification Pipeline screen for the causal_offset_mean_calibration_pilot_2026_07_22 forward candidates on polished_dataset setpoint inputs. The screen compared the two causal offset / mean calibration candidates against the accepted polished-setpoint forward baselines and the recent scalar shape-objective high-water reference.

Execution Summary

Item	Value
Screen	causal_offset_bounded_track2_screen_2026_07_22
Dataset	polished_dataset
Input mode	setpoints
Surface	Fw / forward
Candidate count	5
Curve count	100
Matrix output	output/validation_checks/track2_reference_comparison/2026-07-23-13-07-44__causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_causal_offset_bounded_track2_screen_polished_setpoints_fw
Matrix report	doc/reports/analysis/validation_checks/te_curve_verification_pipeline/2026-07-23-13-08-58_causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_causal_offset_bounded_track2_screen_polished_setpoints_fw_report.md
Reranker output	output/validation_checks/shape_gated_te_curve_reranker/2026-07-23-13-09-03__shape_gated_te_curve_reranker
Reranker report	doc/reports/analysis/te_curve_verification_pipeline/03_cvp_diagnostics/shape_gated_reranker/[2026-07-22]/causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_shape_gated_te_curve_reranker_report.md
Plot count	10
Plot summary	doc/reports/campaign_results/track_2/verification_plots/causal_offset_bounded_track2_screen_polished_setpoints_fw/track2_candidate_curve_plot_summary.yaml

Metric Ranking

Rank	Candidate	Raw MAE [deg]	RMSE [deg]	Mean Error [%]	P95 Error [%]
1	polished_setpoints_periodic_gru_sequence_Fw	0.001837	0.002179	3.750629	8.248098
2	polished_setpoints_periodic_mlp_harmonic_Fw	0.001938	0.002276	3.987765	10.017324
3	shape_objective_periodic_mlp_harmonic_Fw	0.002035	0.002412	4.222146	10.956483
4	causal_offset_mean_periodic_mlp_harmonic_Fw	0.002075	0.002478	4.314268	9.717112
5	causal_offset_mean_gru_sequence_Fw	0.002392	0.002868	5.009841	10.269009

Shape-Gated Decision

Rank	Candidate	Centered MAE [deg]	Shape Pass	Composite	Decision
1	polished_setpoints_periodic_gru_sequence_Fw	0.001483	0.950000	0.007008	recommended
2	polished_setpoints_periodic_mlp_harmonic_Fw	0.001490	0.920000	0.239545	candidate
3	shape_objective_periodic_mlp_harmonic_Fw	0.001578	0.910000	0.419662	candidate
4	causal_offset_mean_periodic_mlp_harmonic_Fw	0.001652	0.920000	0.424363	candidate
5	causal_offset_mean_gru_sequence_Fw	0.002024	0.000000	0.961925	shape gate failed

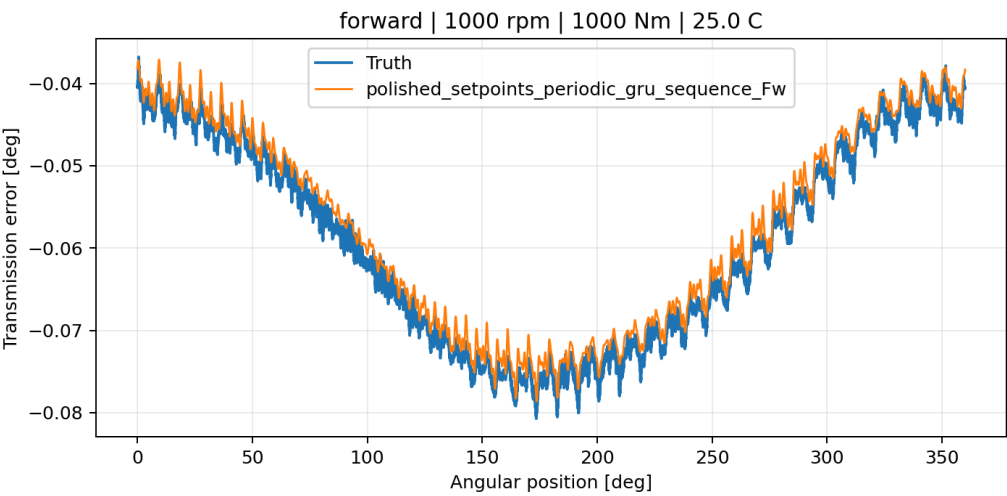
Harmonic Breakdown

Rank	Candidate	FFT Sim.	Amp Err. [%]	Phase Err. [deg]
1	polished_setpoints_periodic_gru_sequence_Fw	0.984971	17.554759	12.349713
2	polished_setpoints_periodic_mlp_harmonic_Fw	0.983739	15.623668	12.951608
3	shape_objective_periodic_mlp_harmonic_Fw	0.983709	18.390567	17.439607
4	causal_offset_mean_periodic_mlp_harmonic_Fw	0.983672	19.654545	18.516624
5	causal_offset_mean_gru_sequence_Fw	0.975176	77.578652	71.215158

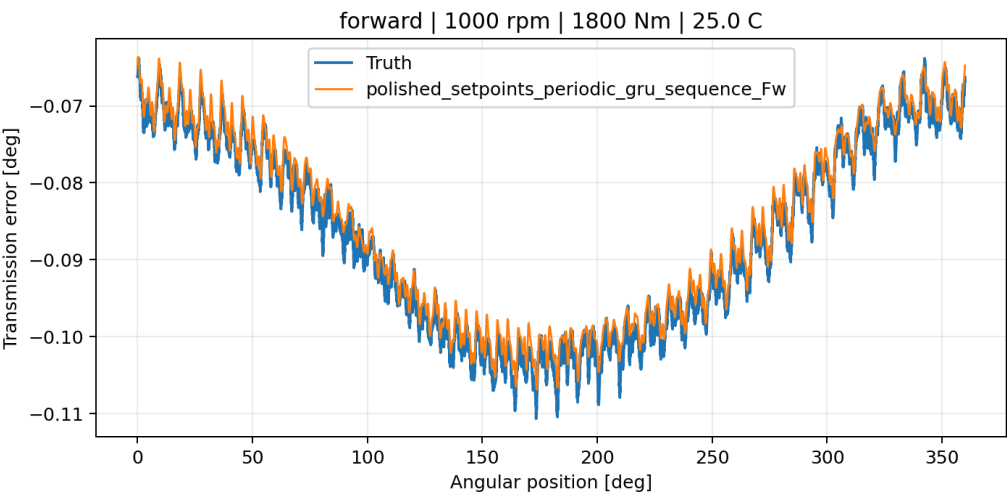
Pilot Graphs

The bounded screen plot contract was backfilled after the closeout finding. The package now contains two measured-versus-predicted TE curve plots for each bounded-screen candidate.

Polished Setpoints Periodic GRU Sequence Fw

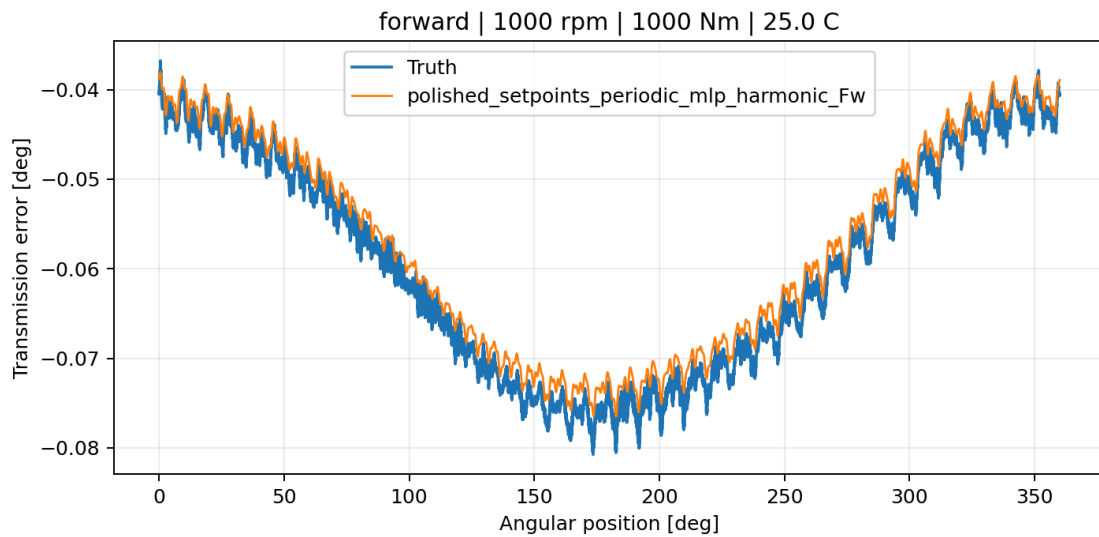


polished_setpoints_periodic_gru_sequence_Fw measured-versus-predicted TE curve

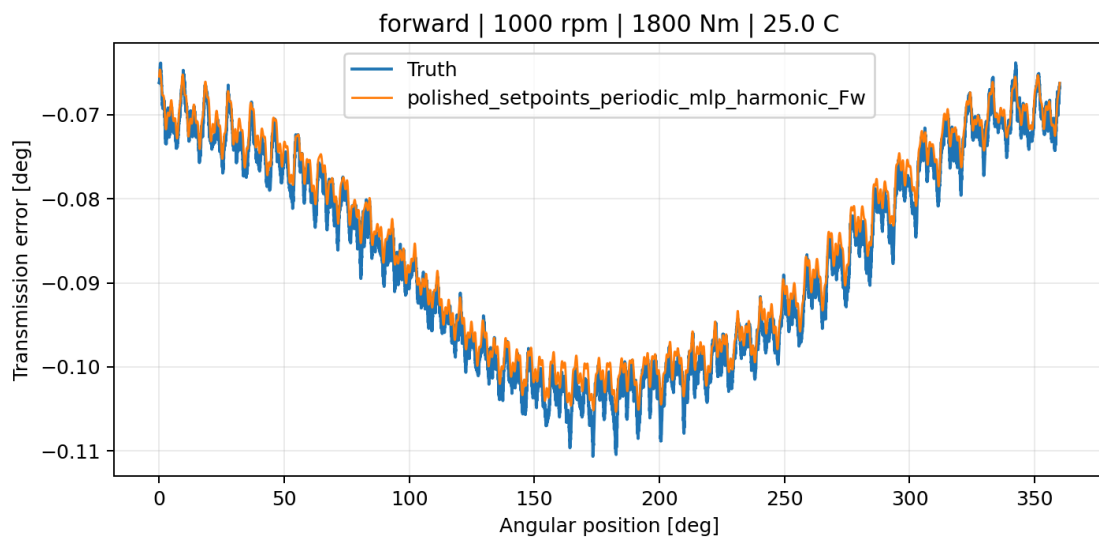


polished_setpoints_periodic_gru_sequence_Fw measured-versus-predicted TE curve

Polished Setpoints Periodic MLP Harmonic Fw

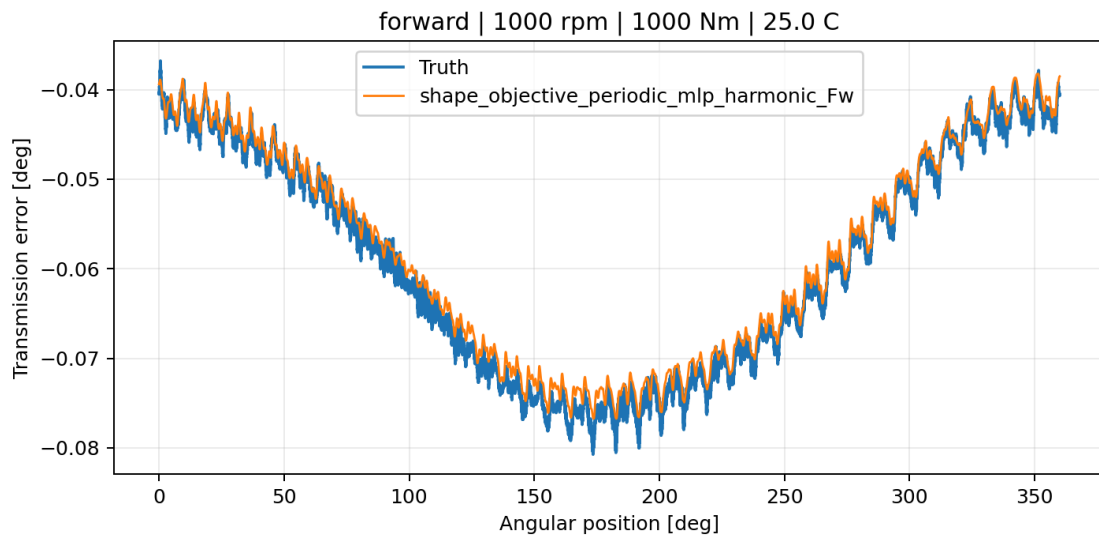


polished_setpoints_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

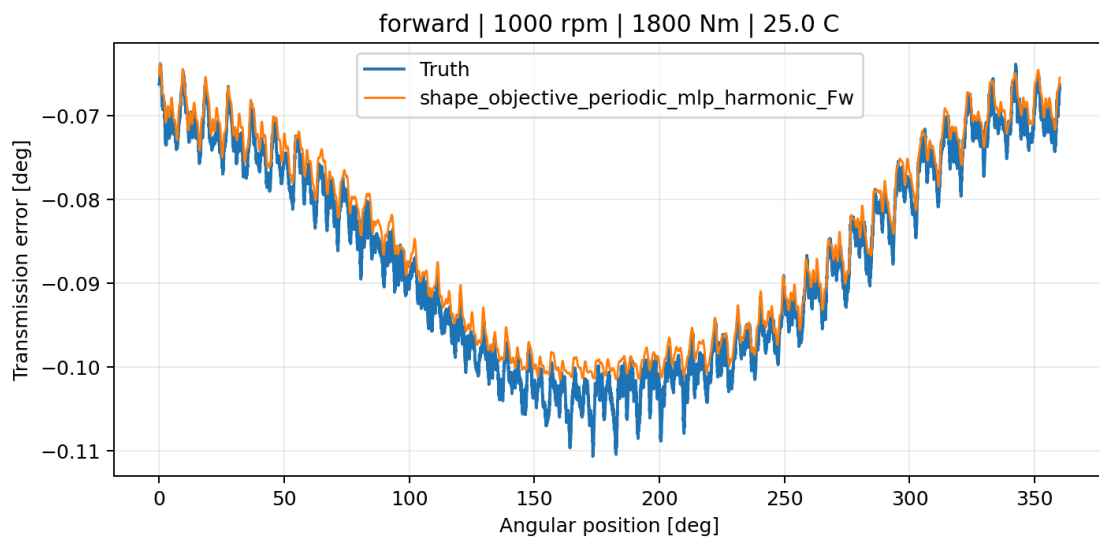


polished_setpoints_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

Shape Objective Periodic MLP Harmonic Fw

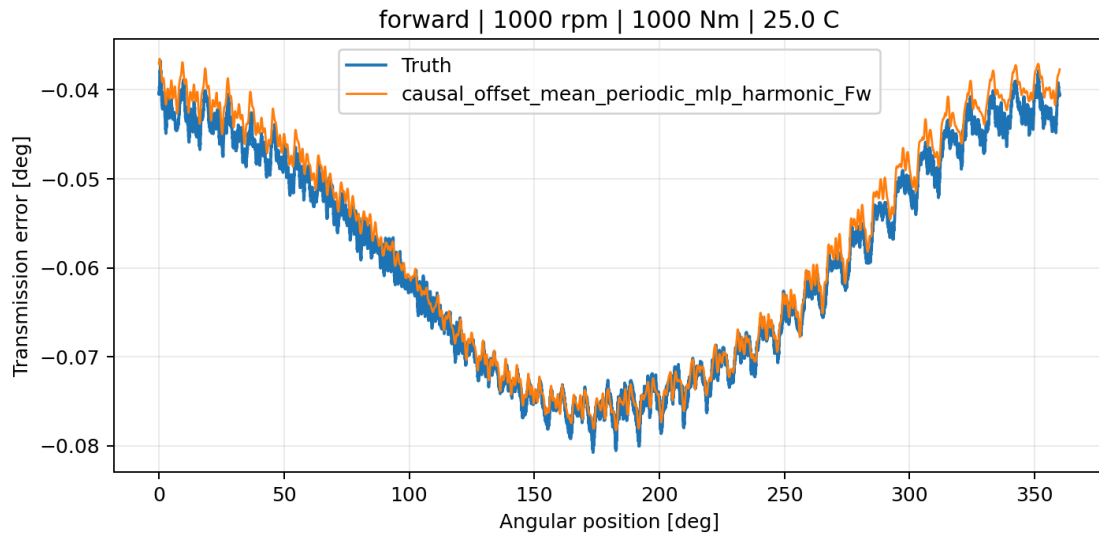


shape_objective_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

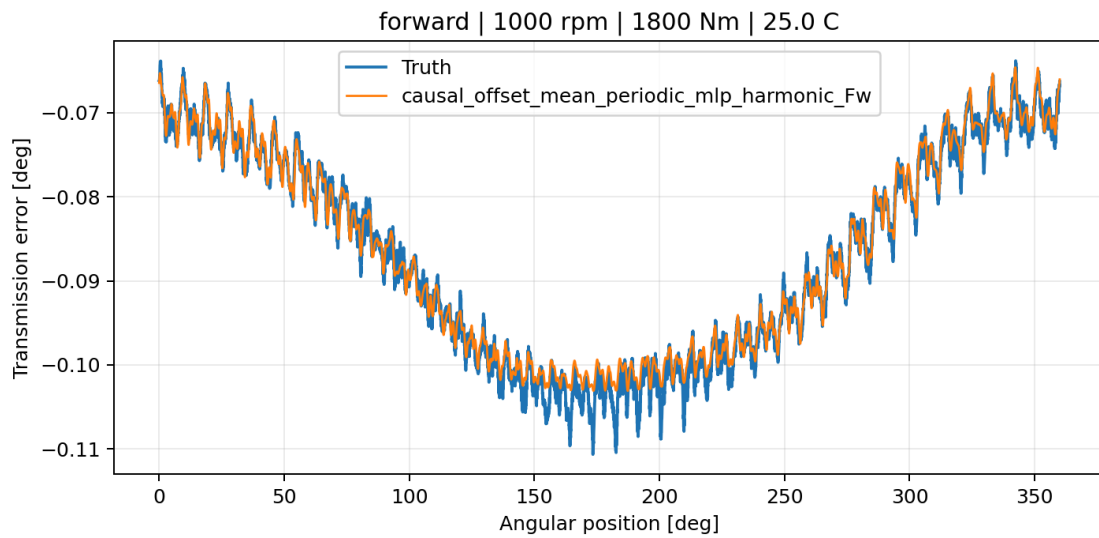


shape_objective_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

Causal Offset Mean Periodic MLP Harmonic Fw

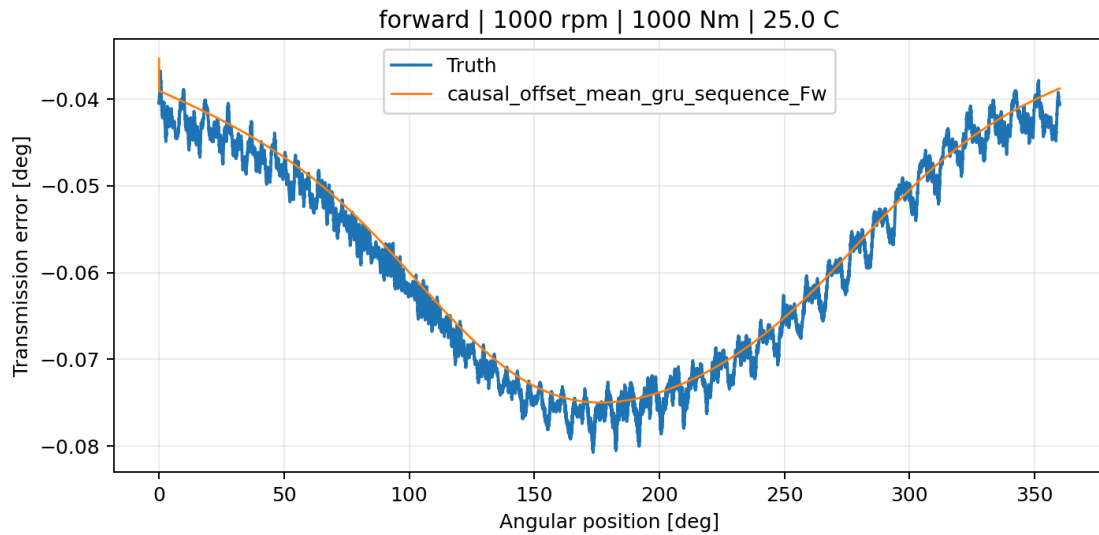


causal_offset_mean_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

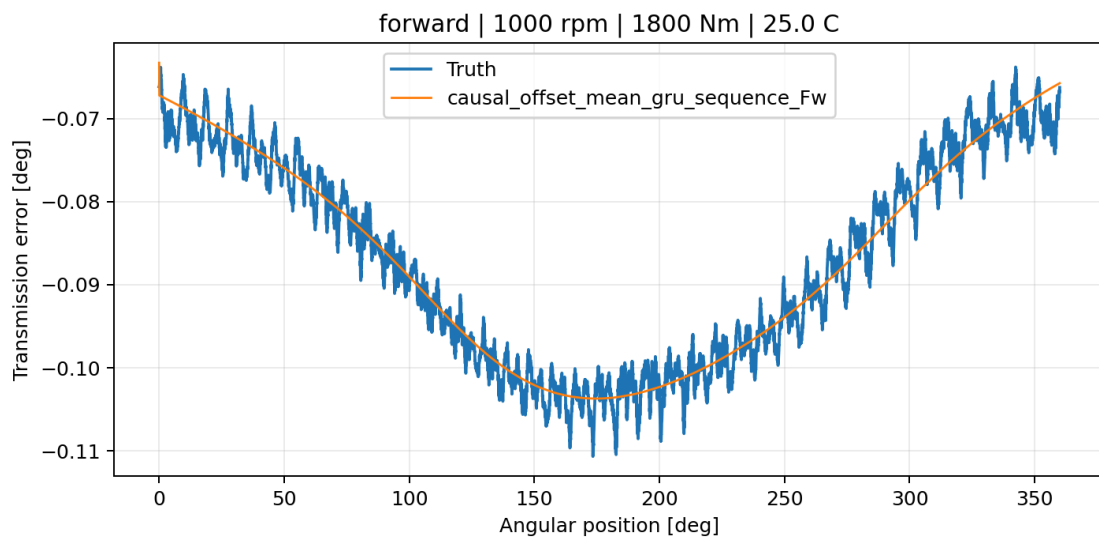


causal_offset_mean_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

Causal Offset Mean GRU Sequence Fw



causal_offset_mean_gru_sequence_Fw measured-versus-predicted TE curve



causal_offset_mean_gru_sequence_Fw measured-versus-predicted TE curve

Interpretation

The bounded screen does not promote either causal offset / mean calibration candidate.

The accepted `polished_setpoints_periodic_gru_sequence_Fw` baseline remains the forward recommendation. It leads on raw MAE, centered MAE, shape pass rate, and composite score.

The non-windowed causal offset MLP remains a valid completed pilot, but it ranks behind the accepted GRU, the non-windowed polished-setpoint MLP baseline, and the prior shape-objective MLP reference. It does not justify expansion from this profile.

The time-windowed causal residual-offset GRU is rejected for this branch. Its shape pass rate is `0.000000`, and the reranker reports failures dominated by harmonic amplitude error, phase error, derivative agreement, FFT similarity, and peak-to-peak error.

Output Quality Closeout

The original matrix summary reported `report_plot_count: 0` with empty preview and report plot lists. The bounded-screen tooling fix has now backfilled the expected Track 2 measured-versus-predicted plot package and made the future bounded launchers run the plot builder explicitly.

The local operator log `.temp/campaign_log.log` also contains very long unwrapped transfer/progress lines and PowerShell host records. The most visible source is interactive `scp` transfer progress emitted during remote sync. This was an operator-output problem, not a model-result change. The permanent launcher/tooling fix now suppresses noisy transfer progress while preserving repository-owned step summaries and failure status.

Decision

Do not promote the causal offset / mean calibration candidates.

Do not expand the causal residual-offset GRU branch from this evidence.

Keep both time-windowed and non-windowed roads as comparison categories for future designs, but close this direct causal-offset mean-calibration profile as non-promoted.

The bounded campaign output contract is now repaired for future analogous screens: remote sync logs should not flood the terminal with unwrapped progress lines, and bounded Track 2 screens should reliably produce the expected measured-versus-predicted TE curve plots when plot generation is part of the operator-facing reporting contract.